

Key Findings

Fraud Detection Analytics Case Study

Executive Summary

This project analyzed financial transaction data to identify fraudulent activities, evaluate customer behavior, and uncover operational risk patterns using SQL, Python, Power BI, and Tableau. Through exploratory data analysis, KPI reporting, and interactive dashboards, the project demonstrates how analytics can support fraud detection and improve decision-making in banking and financial services.

Key Analytical Findings

Transaction Analysis

- Compared fraudulent and legitimate transactions to understand overall fraud distribution.
 - Evaluated transaction volumes and average transaction values to identify abnormal spending behavior.
 - Examined peak transaction periods to detect potential high-risk activity windows.
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Customer Behavior

- Identified customers with repeated suspicious transaction patterns.
 - Analyzed spending behavior to highlight unusual purchasing activity.
 - Segmented customer activity to support risk-based monitoring and investigation.
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Geographic Analysis

- Evaluated transaction activity across different regions.
 - Identified locations with comparatively higher concentrations of suspicious transactions.
 - Examined geographic movement patterns that may indicate potential fraud.
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Time-Based Analysis

- Analyzed transaction trends across daily, weekly, and monthly periods.
 - Investigated recurring fraud patterns over time.
 - Identified transaction periods that warrant closer operational monitoring.
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Business KPIs Developed

The analysis includes key performance indicators that support fraud monitoring and reporting:

- Total Transactions
 - Total Fraud Cases
 - Fraud Rate
 - Average Transaction Value
 - High-Risk Customers
 - Fraud by Region
 - Fraud by Payment Channel
 - Peak Fraud Hours
 - Monthly Fraud Trends
 - Merchant Category Analysis
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Business Insights

The analysis demonstrates how organizations can use data analytics to strengthen fraud prevention by:

- Prioritizing high-risk transactions for investigation.
 - Monitoring customer behavior for unusual activity.
 - Identifying geographic and temporal fraud patterns.
 - Supporting fraud analysts with interactive dashboards and KPI reporting.
 - Enabling faster, data-driven operational decisions.
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Potential Business Applications

The analytical approach demonstrated in this project can be applied across:

- Retail Banking
 - Digital Payments
 - FinTech Platforms
 - Credit Card Operations
 - Insurance Claims Analytics
 - Risk & Compliance Teams
 - Financial Crime Investigation
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Skills Demonstrated

Data Analytics

- Exploratory Data Analysis (EDA)
- Fraud Analytics
- Financial Data Analysis

- Business Intelligence
- KPI Reporting
- Data Storytelling

Technical Skills

- Python
 - SQL
 - Pandas
 - NumPy
 - Matplotlib
 - Power BI
 - Tableau
 - Microsoft Excel
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Conclusion

This project demonstrates an end-to-end analytics workflow that transforms raw transaction data into actionable business insights. By combining Python for data preparation, SQL for analytical querying, and Power BI and Tableau for visualization, the project illustrates practical techniques for identifying fraud patterns, monitoring key performance indicators, and supporting evidence-based decision-making in financial services.